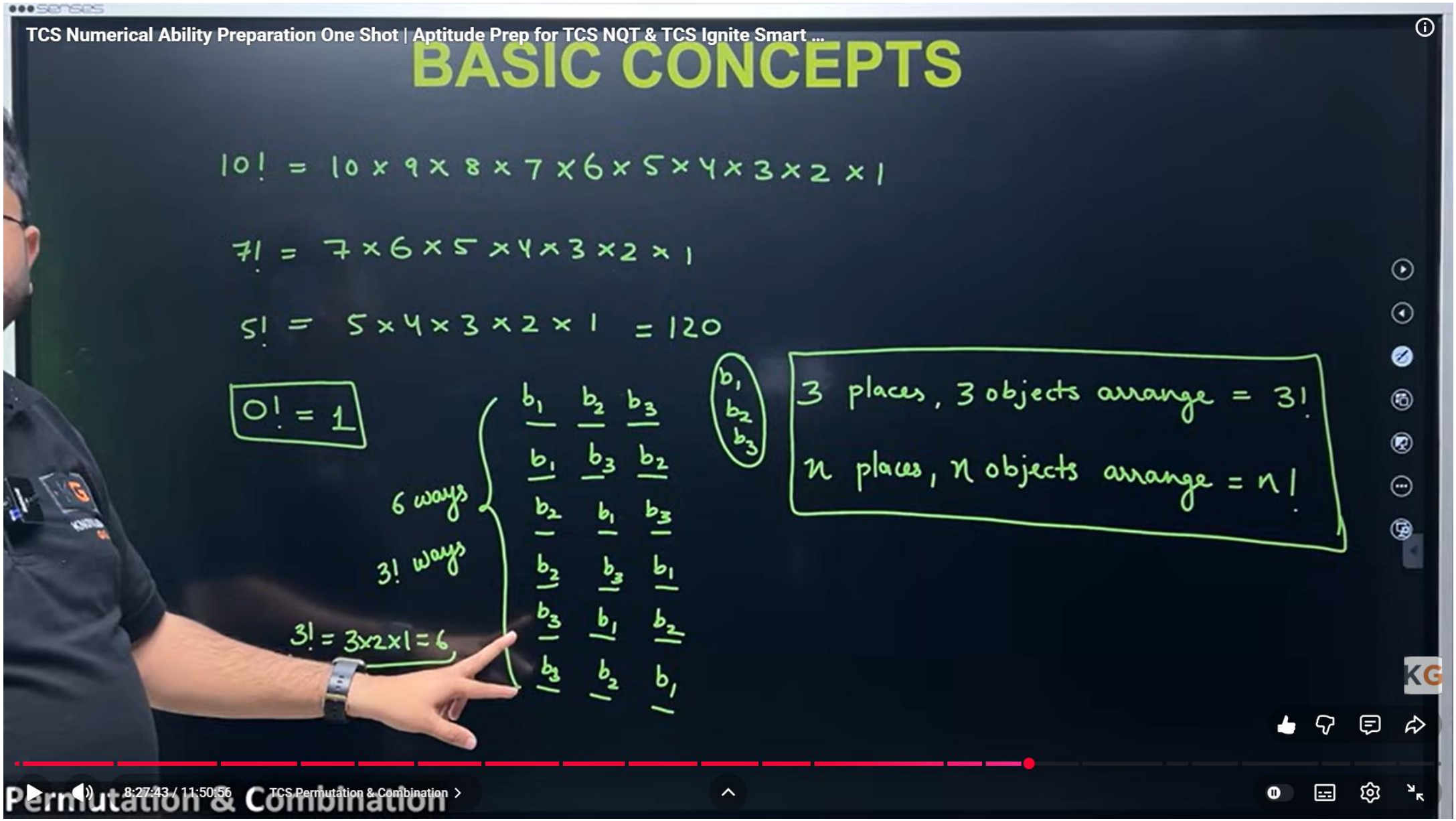
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BASIC CONCEPTS
10! = 10 × 9 × 8 × 7 × 6 × 5 × 4 × 3 × 2 × 1
7! = 7 × 6 × 5 × 4 × 3 × 2 × 1
5! = 5 × 4 × 3 × 2 × 1 = 120
0! = 1
b1 b2 b3
b1 b3 b2
b2 b1 b3
b2 b3 b1
b3 b1 b2
b3 b2 b1
6 ways
3! ways
3! = 3×2×1 = 6
b1 b2 b3
3 places, 3 objects arrange = 3!
n places, n objects arrange = n!
8:27:43 / 11:50:56
TCS Permutation & Combination
Permutation & Combination

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$$^8C_3 = \frac{3 \text{ items}}{3 \text{ items}} = \frac{8 \times 7 \times 6}{1 \times 2 \times 3} = 56$$

$$\boxed{^8C_6 = {}^8C_2}$$

$$^8C_6 = \frac{6 \text{ items}}{6 \text{ items}} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3}{1 \times 2 \times 3 \times 4 \times 5 \times 6} = \frac{\overset{4}{8} \times 7}{2} = 28$$

$$^8C_2 = \frac{2 \text{ items}}{2 \text{ items}} = \frac{\overset{4}{8} \times 7}{1 \times 2} = 28$$

$$\boxed{^nC_r = {}^nC_{n-r}}$$

8 people → 6 people $^8C_6$
→ 2 people $^8C_2$ } same

n people → r people
→ n-r people } same

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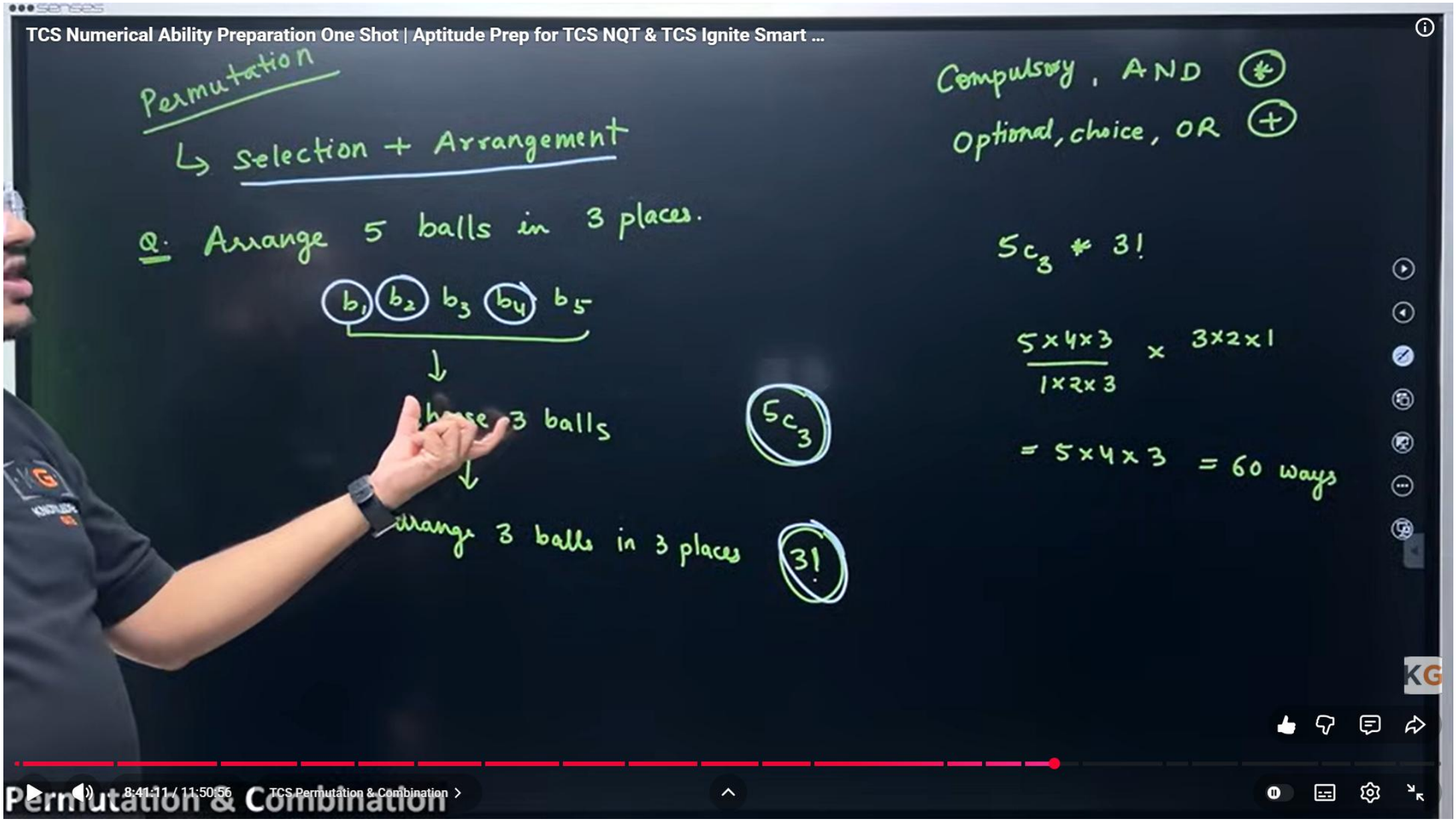
Permutation & Combination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Permutation
↳ Selection + Arrangement
Q. Arrange 5 balls in 3 places.
b1 b2 b3 b4 b5
choose 3 balls
5C3
arrange 3 balls in 3 places
3!
Compulsory, AND *
Optional, choice, OR +
5C3 * 3!
5×4×3 / 1×2×3 × 3×2×1
= 5×4×3 = 60 ways
8:41:11 / 11:50:56
TCS Permutation & Combination
Permutation & Combination

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Q1: If we have 10 people and 4 chairs, what are the number of ways to seat them?

A) 5040 ✓
B) 4880
C) 3600
D) 6890

M1

$$^{10}P_4 = \frac{10!}{(10-4)!} = \frac{10 \times 9 \times 8 \times 7 \times \cancel{6!}}{\cancel{6!}}$$

$6 \times 5 \times 4 \times 3 \times 2 \times 1$

$$^{n}P_r = \frac{n!}{(n-r)!}$$

$= 10 \times 9 \times 8 \times 7 = 5040$ ways

M2 $^{10}C_4 * 4! = 5040$ ways

M3

$P_2 \quad P_4 \quad P_8 \quad P_7$

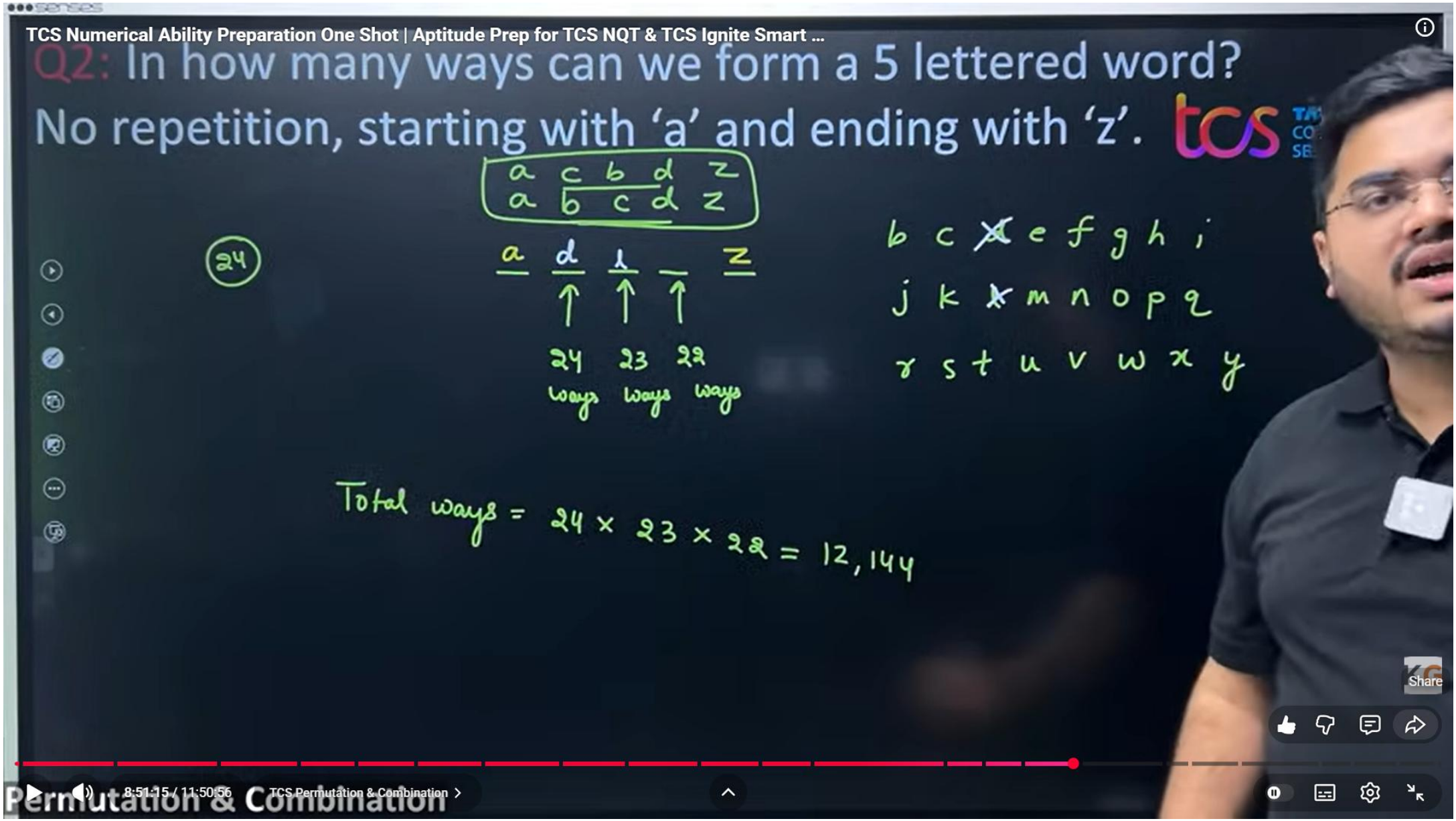
10 ways 9 ways 8 ways 7 ways

$\cancel{P_1}\ \cancel{P_2}\ P_3\ \cancel{P_4}\ P_5$
$P_6\ P_7\ \cancel{P_8}\ P_9\ P_{10}$

total $= 10 \times 9 \times 8 \times 7$
$= 5040$ ways

Permutation & Combination

8:47:50 / 11:50:56 TCS Permutation & Combination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Q2: In how many ways can we form a 5 lettered word?
No repetition, starting with 'a' and ending with 'z'.
a c b d z
a b c d z
24
a d _ _ z
24 ways
23 ways
22 ways
b c ~~d~~ e f g h i
j k ~~l~~ m n o p q
r s t u v w x y
Total ways = 24 × 23 × 22 = 12,144
8:51:15 / 11:50:56
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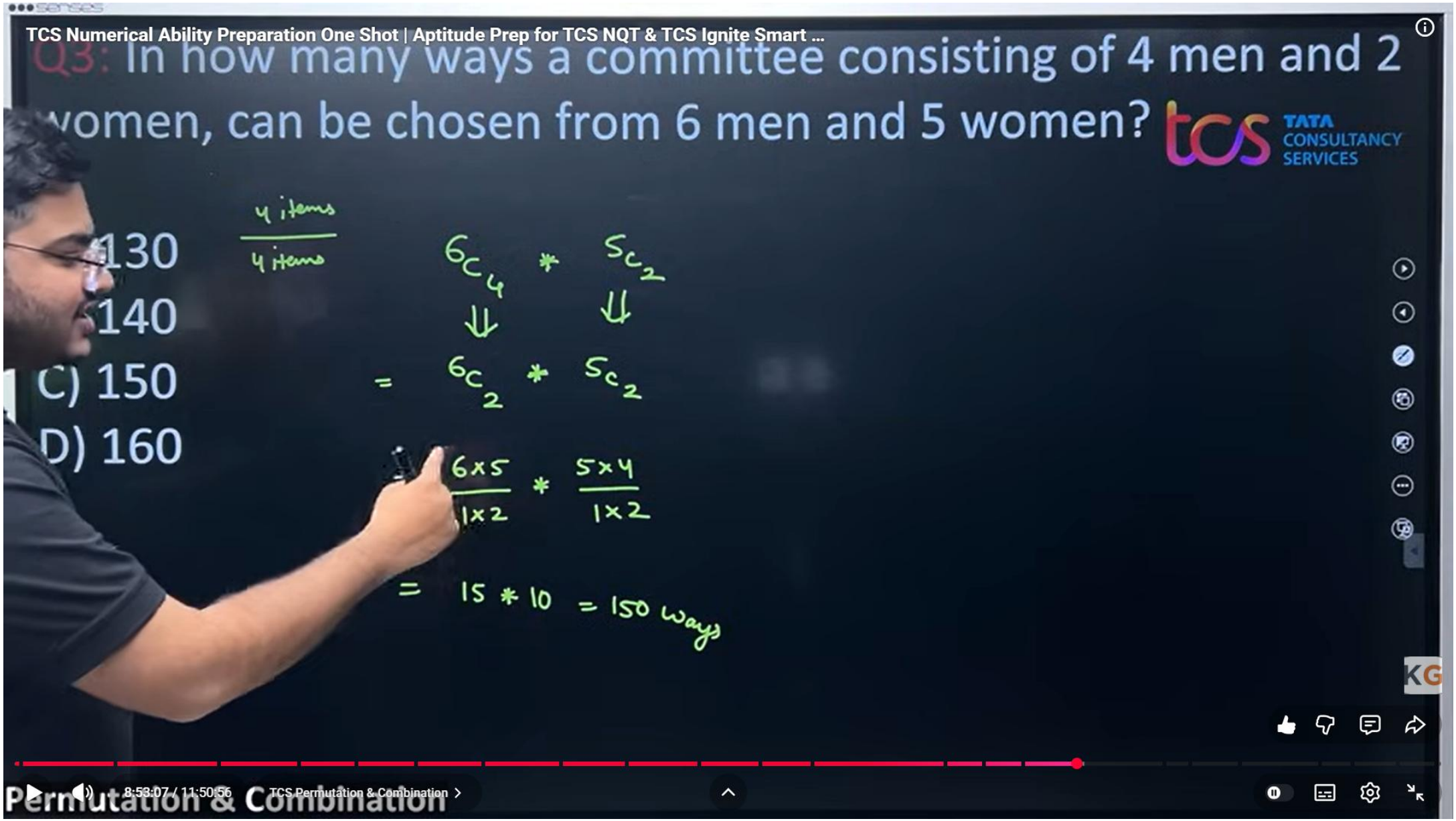
Permutation & Combination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Q3: In how many ways a committee consisting of 4 men and 2 women, can be chosen from 6 men and 5 women?
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130
140
C) 150
D) 160
4 items
4 items
6C4 * 5C2
= 6C2 * 5C2
6x5 / 1x2 * 5x4 / 1x2
= 15 * 10 = 150 ways
8:53:07 / 11:50:56
TCS Permutation & Combination
Permutation & Combination